

Contact

(786)-804-7278 (Work)
alexeygallegomartinez@gmail.com

www.linkedin.com/in/alexey-martinez (LinkedIn)

Top Skills

System Architecture
Business Process Automation
Cloud Services

Certifications

Introduction to HTML, CSS, & JavaScript
Introduction to Software Engineering
Supervised Machine Learning: Regression and Classification
Programming with JavaScript
Introduction to Front-End Development

Alexey Martinez

Systems Builder | Software Engineer | AI & Business Systems |
Technical Leadership | Automation
Ann Arbor, Michigan, United States

Summary

I believe engineering is ultimately about solving problems that matter.

My journey started long before software development. After immigrating to the United States, I worked my way through engineering school while helping support my family, balancing multiple jobs, academics, and personal responsibilities. Those experiences taught me resilience, adaptability, and the ability to learn quickly under pressure.

Throughout my career, I have repeatedly stepped into unfamiliar challenges, learned whatever was necessary to solve them, and built solutions that create measurable business impact. What began with mechanical engineering, robotics, and industrial automation evolved into full-stack software development, ERP systems, AI integration, cloud infrastructure, and business operations.

Today, I lead development efforts for a company-wide ERP platform while also driving architecture, infrastructure, and technical decision-making across multiple business-critical systems. I have architected internal business platforms used daily by over 100 employees, implemented AI-powered workflows that significantly reduce administrative effort, developed industrial monitoring systems that improved process reliability from 80% to 99%, and led development of a patent-pending robotic solution deployed across manufacturing facilities throughout the United States and Canada.

What excites me most is identifying inefficient processes, understanding the underlying business problem, and building scalable systems that help organizations operate better. Whether the solution involves software, automation, AI, infrastructure, or product development, my approach remains the same: take ownership, learn quickly, and deliver results.

I am passionate about technology, entrepreneurship, continuous learning, and helping organizations grow through engineering, innovation, and operational excellence.

Experience

Encore Automation

Systems Engineer (Full Stack Software Engineer)

July 2021 - Present (5 years)

Auburn Hills, Michigan, United States

As a Systems & Software Engineer at Encore Automation, I work at the intersection of software development, business systems, industrial automation, and product development. I currently lead development efforts for a company-wide ERP platform that supports engineering, purchasing, finance, receiving, and project management operations while defining architecture standards, conducting code reviews, mentoring engineers, and driving technical direction across the platform.

Beyond ERP development, I independently architected and built an internal operations platform used daily by over 100 employees, replacing manual spreadsheet-based processes with modern workflows for timesheets, expenses, scheduling, approvals, reporting, and notifications. These improvements increased timesheet compliance from approximately 70% to 95% and expense submission compliance from approximately 50% to 90%.

I have also introduced AI-powered workflows, including automated receipt analysis using OpenAI Vision, reducing administrative review effort from roughly one business day to one to two hours. In addition, I own the infrastructure supporting these systems, including Ubuntu servers, Docker deployments, CI/CD pipelines, and production environments maintaining 99%+ uptime.

Outside of software, I have led development of a patent-pending robotic sealing solution deployed across automotive manufacturing facilities throughout the United States and Canada, reducing cycle times by approximately 50%, improving quality, and generating significant revenue for the business. I also developed industrial monitoring applications that interface

directly with FANUC robots, improving process reliability from approximately 80% to 99%.

Throughout my time at Encore, I have mentored engineering interns and junior developers while consistently identifying opportunities to improve business operations through technology, automation, and scalable systems.

uBreakiFix

Manager / Lead Technician

July 2020 - July 2021 (1 year 1 month)

Miami, Florida, United States

Led daily operations for a team of technicians and customer service personnel, improving workflow efficiency and increasing store revenue by 50%.

Diagnosed and repaired complex electronic and hardware systems for Apple, Samsung, and Google devices.

Managed inventory, procurement, and technical service operations while maintaining high customer satisfaction.

GE Appliances, a Haier company

Technology Co-Op

January 2020 - June 2020 (6 months)

Louisville, Kentucky, United States

Supported development and testing of residential HVAC and ductless air conditioning systems.

Performed performance analysis, validation testing, and troubleshooting activities to improve product efficiency and reliability.

Designed, built, and tested mechanical components while ensuring system-level requirements were achieved.

Developed engineering tools and dashboards to improve project visibility and cross-functional coordination.

Florida International University

Statics Learning Assistant

January 2019 - December 2019 (1 year)

Taught students to apply Calculus and Physics concepts to solve complex engineering problems.

Collaborated with the professor to guide students in class, enhancing their understanding of engineering fundamentals.

Conducted tutoring sessions to assist students and answer course-related questions.

Education

Florida International University - College of Engineering & Computing
Bachelor of Science - BS, Mechanical Engineering (Robotics
Focus) · (2018 - May 2021)